

1 Establishment and Remodeling of the
2 *Mycobacterium tuberculosis* 3D Genome
3 Orchestrate Context-Dependent Transcriptional
4 Program

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Abstract

The sophisticated three-dimensional (3D) genome architecture has emerged as a fundamental organizational feature in prokaryotes and orchestrates gene transcription. However, the mechanisms governing its formation, remodeling, and coordination with transcription remain largely elusive. We leveraged *Mycobacterium tuberculosis* as a prokaryotic model and integrated Hi-C, RNA-seq, ChIP-seq, and protein interactome data to construct a comprehensive multi-layer protein–DNA interaction network. Within this framework, the conserved nucleoid-associated protein NapM bridges DNA–protein interactions and functions as a key genome organizer. Its absence impairs *M. tuberculosis* chromosome architecture and disrupts synchronized operon interactions and their underlying transcriptional coordination. Furthermore, the *M. tuberculosis* 3D genome exhibits pronounced plasticity and undergoes dynamic remodeling under diverse stress conditions to coordinate transcriptional reprogramming. Importantly, Shannon entropy and Moran’s I analyses revealed that NapM is essential for maintaining chromatin architectural plasticity and enabling adaptive remodeling under stress. Loss of NapM consequently induces transcriptional rigidity and disrupts dormancy establishment. Together, our study elucidates fundamental principles governing prokaryotic 3D genome formation, remodeling, and coordinated transcriptional regulation.

Keywords: chromosome plasticity, NapM, stress adaptation, transcriptional regulation

1 Introduction

The 3D organization of genome is an evolutionarily conserved architectural code that orchestrates DNA condensation, replication, segregation, and transcriptional control across all domains of life. In eukaryotes, chromatin is elegantly partitioned into hierarchical structural units such as chromatin loops and topologically associating domains (TADs), stabilized by a set of sophisticated architectural formation machineries including CTCF and cohesin[1–7]. These structures govern enhancer–promoter communication to define dynamic spatial neighborhoods, thereby ensuring transcriptional efficiency, specificity and regulatory insulation in response to diverse physiological processes[8–12]. Recent high-content functional screens have further uncovered a diverse set of genes that collectively contribute to the establishment, stabilization, or remodeling of higher-order genome folding, such as cohesin regulators and loaders (NIPBL/SCC2, WAPL, and PDS5), condensin complexes, architectural transcription factors including YY1 and TFIIC, components of the Mediator complex, RNA polymerase II (Pol II), and topoisomerases. These factors constitute a multi-layered regulatory machinery, in which dynamic protein–protein and protein–DNA interactions act in concert to sustain and remodel higher-order genome folding across developmental and environmental contexts[13, 14]. While a large body of evidence has elucidated the establishment and dynamic remodeling of the 3D genome in eukaryotes, the machinery underlying 3D genome folding in prokaryotes and the embedded transcriptional codes remain elusive.

Advances in chromosome conformation capture technologies have revealed that prokaryotic genomes also exhibit highly ordered and dynamic folding patterns. chromosome interaction domains (CIDs) and loop-like structures have been identified in model species such as *Escherichia coli*, *Bacillus subtilis*, and *Caulobacter crescentus*, reflecting a compartmentalized nucleoid organization [15–21]. Emerging evidence has revealed that several structural maintenance of chromosome (SMC) complexes and nucleoid-associated proteins (NAPs) cooperatively orchestrate chromosome compaction, segregation, and gene transcription [22, 23]. The MatP/matS system at the *E. coli* terminus exemplifies a specific protein–DNA bridging mechanism that defines chromosome domain boundaries [24], while NAPs such as histone-like nucleoid structuring protein (H-NS), factor for inversion stimulation (Fis), integration host factor (IHF), and heat-unstable nucleoid protein (HU) modulate local DNA stiffness and transcriptional accessibility [25]. However, it remains largely unknown which prokaryotic proteins are involved, and how they act cooperatively with each other and their specific DNA targets to establish and remodel chromosome architecture to orchestrate dynamic transcriptional programs in response to diverse environmental stimuli.

Emerging evidence shows that the architecture of the bacterial chromosome is tightly coupled to transcriptional programming. In *E. coli* and *Bacillus subtilis*, transcriptionally active regions frequently coincide with CID boundaries, indicating that gene transcription and 3D folding might be reciprocally regulated [15, 20, 23, 26]. These transcription-induced topological changes, together with SMC and NAP activity, dynamically remodel nucleoid architecture and contribute to chromosome domain formation. For example, H-NS-mediated nucleoprotein structures in enteric bacteria are critical for selective silencing AT-rich horizontally acquired genes and can directly modulate transcription by increasing RNA polymerase pausing and promoting Rho-dependent termination. Likewise, in mycobacteria, NAPs, including Lsr2 and EspR, couple chromosome organization to broad transcriptional control, including virulence-associated loci [27–30].

While increasing investigations have been dedicated to the bacterial chromosome, it is crucial to note with caution that most model bacteria possess remarkably rapid replication and very short generation times. This significantly hinders precise delineation and detailed temporal mapping of chromosome architecture during a rapid cell cycle [31]. In this scenario, *Mycobacterium tuberculosis* exhibits a markedly slower division rate (18–24 h per division, in comparison to 20 min per division for *Escherichia coli*) [32], providing a substantially extended window for capturing dynamic nucleoid states and 3D structural rearrangements. Therefore, we employed *Mycobacterium tuberculosis* as an ideal prokaryotic model to elucidate the establishment and remodeling of the prokaryotic 3D genome and its role in transcriptional regulation during environmental adaptation. By integrating Hi-C, RNA-seq, ChIP-seq, and systematic analyses of protein–protein and protein–DNA interactions, we reconstruct a comprehensive multilayer interaction network encompassing DNA–DNA, DNA–protein, and protein–protein interactions. By leveraging this integrative dataset, we systematically screened the architectural machinery underlying the prokaryotic 3D genome, dissected the cooperative principles governing chromosome organization, and elucidated its role in transcription regulation.

117 2 Results and Discussion

118 The *M. tuberculosis* 3D genome and its underlying 119 transcriptional domains

120 Leveraging the slow growth rate of *M. tuberculosis*, we employed it as a model system
121 to elucidate the fine-scale spatial organization of the prokaryotic genome. We optimized
122 the protocol for high-resolution Hi-C in prokaryotes, which generated high-quality Hi-
123 C maps with high reproducibility across four biological replicates (Extended Data
124 Fig. 1a–c). The genome-wide contact map revealed a pronounced ori–ter organization,
125 consistent with a longitudinally folded circular chromosome (Fig. 1a). Reconstruction
126 of the chromosome model further confirmed that the replication origin (ori) and
127 terminus (ter) occupy opposite ends of the nucleoid, indicative of an ordered axial
128 chromosome arrangement (Fig. 1b). At the compartment level, saddle plot analy-
129 sis demonstrated clear A/B compartmentalization, with preferential A–A and B–B
130 interactions over A–B contacts, indicating higher-order spatial segregation of the
131 chromosome (Fig. 1c). In addition, genomic profiling of GC skew and GC content
132 enabled precise identification of the replication origin and terminus, providing indepen-
133 dent sequence-based anchors that support the reconstructed chromosome architecture
134 (Extended Data Fig. 1d).

135 Integration of multiple omics layers—including CIDs, directionality index (DI),
136 RNA-seq expression profiles, and loop annotations—revealed a close spatial corre-
137 spondence between chromosome architecture and transcriptional activity (Fig. 1d). In
138 concert with this highly organized chromosome architecture, genes located within the
139 same Hi-C–defined CIDs exhibited significantly increased gene co-expression level (or
140 reduced expression variance) compared with randomized genomic controls, indicat-
141 ing coordinated transcription within the same structural domains (Fig. 1e). Similarly,
142 gene pairs located at the anchors of long-range loops, yet residing in different operons,
143 exhibited a significantly higher co-expression level than that of the distance-matched
144 random pairs, suggesting enhanced co-regulation of gene transcription associated with
145 spatial proximity (Fig. 1f).

146 To further examine transcriptional coordination at the operon level, we con-
147 structed an operon–operon interaction matrix according to Hi-C contact frequencies
148 for 749 annotated operons (Fig. 1g). Based on this matrix, we identified a subset
149 of Cooperative operon hubs (COHs), defined as operon pairs exhibiting strong chro-
150 mosome interaction strength and tightly synchronized transcription (Fig. 1h–j). For
151 example, representative pairs in COHs, including *Rv3001–Rv3049c* operon pair and
152 *Rv2616_opn–Rv2688c_opn* operon pair, exhibited strong Hi-C interactions and tightly
153 coordinated expression patterns at the operon level (Fig. 1k,l). 3D reconstruction also
154 confirmed their close spatial proximity within the nucleoid, supporting the structural
155 basis for operon co-regulation.

156 Functional enrichment analysis of genes encoded by COH-associated operons
157 revealed significant overrepresentation of pathways related to transport, metabolism,
158 and stress-associated processes. Gene Ontology analysis highlighted enrichment in
159 biological processes such as branched-chain amino acid catabolism, transmembrane
160 transport, and carbohydrate metabolism, as well as molecular functions linked to

161 transporter activity (Fig. 1m). Consistently, KEGG pathway analysis also identi-
 162 fied enrichment of ABC transporters and central metabolic pathways, suggesting
 163 that COHs preferentially organize functionally related operons involved in core
 164 physiological processes (Fig. 1n).

165 **NapM acts as a key organizer within multilayered** 166 **DNA–protein interaction networks orchestrating chromosome** 167 **architecture**

168 To elucidate the emergence of the 3D genome in *M. tuberculosis*, we constructed a
 169 comprehensive omics integrated network encompassing DNA–DNA contacts, DNA–
 170 protein binding, and a protein–protein interaction(PPI) network based on our Hi-C
 171 dataset and previous ChIP and PPI dataset[33–35] (Fig. 2a,b). This network encom-
 172 passes 2,992 proteins, 883 genomic bins, 8,043 protein–protein interactions, 17,223
 173 protein–DNA events, and 71 DNA–DNA loops (Extended Data Fig. 2a). Each loop
 174 was annotated for direct DNA binding proteins based on ChIP occupancy and bridg-
 175 ing scaffold protein complexes based on PPI network. These data revealed that the
 176 majority of loops (95.8%) shared at least one common protein directly bound to
 177 both anchors. In contrast, CID boundaries typically showed co-binding by multiple
 178 architectural proteins. These findings suggest that most chromosome structures in
 179 *M. tuberculosis* are protein-anchored and might be stabilized through local protein
 180 complexes. Conceptually, the coupling of DNA and protein interactions may define a
 181 hierarchical framework that stabilizes the bacterial chromosome (Fig. 2c).

182 To further dissect the architectural roles of proteins within this integrated net-
 183 work, we extracted protein interaction subnetworks associated with CID boundaries
 184 and loops based on ChIP-seq occupancy. Network analysis revealed that *Rv0047c*
 185 (NapM) consistently occupies a central position in both CID boundary-associated and
 186 loop-associated protein interaction networks, indicating its prominent role in orga-
 187 nizing higher-order chromosome structures (Fig. 2d,e). Moreover, when constructing
 188 a DNA–protein interaction network at the operon level, NapM maintained a pivotal
 189 central role, linking multiple operon-associated interactions(Extended Data Fig. 2b).
 190 Quantitative analysis further showed that, compared with the proteome-wide aver-
 191 age of DNA-binding proteins, NapM is significantly enriched at CID boundaries, loop
 192 anchors, and operon-associated regions, and exhibits increased connectivity within the
 193 PPI network (Fig. 2f). Together, these results indicated NapM as a central architec-
 194 tural factor across multiple layers of genome organization, supporting its role as a key
 195 integrator of DNA–DNA, DNA–protein, and protein–protein interactions.

196 Notably, cross-species analysis demonstrated that NapM homologs are broadly dis-
 197 tributed across bacteria, spanning 303 genera, 2,348 species, and 2,455 orthologous
 198 proteins (Fig. 2g). The widespread phylogenetic distribution of NapM across diverse
 199 bacterial lineages, as summarized by taxonomic categories shown in the correspond-
 200 ing legend (Extended Data Fig. 2e), implies that NapM may play a conserved role in
 201 establishing chromosome architecture. At the sequence level, multiple alignment anal-
 202 yses revealed substantial amino-acid conservation between *M. tuberculosis* NapM and
 203 homologs from both closely related mycobacterial species and more distantly related
 204 taxa, including *S. terrae* and *G. paraffinivorans* (Fig. 2h). Moreover, conserved regions

205 were also observed in comparisons with phylogenetically distant organisms such as
206 *B. subtilis* and *S. criceti* (Extended Data Fig. 2d), indicating ancient evolutionary
207 conservation of core sequence features.

208 To further investigate the role of *napM* in 3D genome establishment, we generated
209 a *napM* knockout strain and compared its genome structure with that of the wild-type
210 (WT). Hi-C analysis revealed that the *napM*-KO strain showed exhibited long-range
211 contacts (Extended Data Fig. 2c). Scalogram analysis, which reflects the distribution
212 of contact distances, showed an expansion of blue regions in the *napM*-KO strain,
213 indicating broader interaction ranges and reduced chromosome compaction (Fig. 2i,j).
214 Consistently, the reconstructed 3D genome model displayed expanded chromosomal
215 arms (Fig. 2k), supporting a globally decondensed chromosome conformation upon
216 NapM depletion.

217 Transcriptomic profiling revealed differential gene expression between WT and
218 *napM*-KO (Extended Data Fig. 3a). Subsequent integrative analysis of ChIP-seq
219 and RNA-seq datasets further showed that, among 634 genes associated with NapM
220 binding peaks, 143 exhibited significant transcriptional changes upon *napM* dele-
221 tion (Extended Data Fig. 3c,d). For example, *Rv2629* was markedly upregulated,
222 whereas *Rv1195* was significantly downregulated in the *napM*-KO strain(Extended
223 Data Fig. 3e,f), indicating that NapM binding can exert both activating and repressive
224 effects on gene expression. To further link transcriptional changes with chromosome
225 architecture, we compared chromosome interaction changes between the regions with
226 differentially expressed genes and transcriptionally stable genes. Notably, gene pairs
227 involving differentially expressed genes exhibited significantly greater changes in chro-
228 mosome interaction strength compared with stable gene pairs ($P = 0.0013$)(Extended
229 Data Fig. 3b), suggesting that NapM-associated transcriptional reprogramming might
230 be coupled to structural reorganization of the genome. Together, these results demon-
231 strate that NapM binding is tightly coupled to transcriptional regulation and that its
232 loss leads to coordinated changes in both gene expression and chromosome interaction
233 patterns.

234 Notably, transcriptional coordination between different operons within COHs was
235 markedly disrupted in the *napM*-KO strain. While operons within representative
236 COHs displayed nearly perfect co-expression in WT (Pearson's $r = 0.9967$), this cor-
237 relation dropped to $r = 0.6214$ following *napM* deletion. This substantial reduction
238 in transcriptional synchrony underscores the critical role of NapM in maintaining
239 long-range operon coordination through genome architectural organization (Fig. 2l).

240 Coordinated chromosome architecture remodeling and 241 transcriptional reprogramming under stress conditions

242 To further elucidate the coordination between genome structure and gene tran-
243 scription, we investigated chromosome architecture remodeling and transcriptional
244 reprogramming in *M. tuberculosis* under stress conditions, including hypoxia and
245 latency (Fig. 3a,b; Extended Data Fig. 4). Both stress conditions induced pronounced
246 alterations in chromosome contact patterns, albeit with distinct structural signatures.

247 These divergent architectural responses were further supported by multi-scale anal-
248 yses of contact probability decay (Fig. 3c). Directionality index (DI) profiles also

revealed extensive remodeling of CIDs under both stress conditions (Fig. 3d), with multiple domain boundaries shifting or merging, indicating dynamic re-partitioning of chromosome domains in response to diverse environmental stresses.

We next reconstructed chromosome structures from Hi-C data to capture these stress-associated topological transitions (Fig. 3e). The WT genome adopted a characteristic ring-like configuration with well-separated ori and ter regions. In contrast, both hypoxic and latent bacteria exhibited reduced spatial separation between the origin (ori) and terminus (ter), indicating enhanced longitudinal chromosome compaction. This effect was more pronounced under latent conditions (Fig. 3f). Together, these observations demonstrate that *M. tuberculosis* undergoes large-scale and plastic remodeling of its chromosome architecture, transitioning between compacted and more decondensed topological states in response to environmental conditions.

Next, we further investigated how the stress-induced structural remodeling is accompanied by coordinated transcriptional reprogramming. Differential expression analysis identified 1,507 genes altered under hypoxia and 1,922 genes under latent conditions relative to WT, respectively (Fig. 3g,h). Under hypoxia, DEGs were enriched for pathways associated with ribosome biogenesis and oxidative phosphorylation (Extended Data Fig. 6a), consistent with rapid suppression of translational capacity and respiratory activity. In contrast, DEGs in the latent state were predominantly enriched for metabolic pathways, including carbohydrate derivative metabolism, organophosphate metabolism, and biosynthetic processes (Extended Data Fig. 6b), reflecting extensive metabolic reprogramming associated with dormancy.

To further investigate the relationship between chromosome reorganization and transcriptional changes, we compared the amplitudes of chromosome interaction changes between differentially expressed gene pairs (DEG–DEG) and transcriptionally stable gene pairs (Stable–Stable) under stress conditions. The magnitude of interaction change was quantified as the absolute $\log_2$ fold change in contact intensity relative to WT. Under hypoxic conditions, DEG–DEG pairs exhibited significantly greater changes in chromosome interaction strength compared with Stable–Stable pairs (one-sided Mann–Whitney U test, $P = 0.0052$) (Fig. 3i). A similar trend was observed in the latent model, where DEG–DEG pairs again showed significantly larger interaction changes than Stable–Stable pairs ($P = 0.0010$) (Fig. 3j). These results indicate that transcriptionally responsive genes are preferentially associated with structural remodeling.

Importantly, integrating Hi-C and RNA-seq data revealed that stress-induced chromosome remodeling drives extensive restructuring of operon-level transcriptional coordination (OTC). The COHs identified under WT conditions lost their transcriptional synchrony under both hypoxic and latent states, with the Pearson correlation coefficient decreasing sharply from 0.9967 in WT to 0.2075 under hypoxia and 0.3188 under latency, respectively (Fig. 3k,l). We next reconstructed condition-specific operon interaction matrices for the hypoxic and latent states based on their respective Hi-C data and derived new sets of COHs under each stress condition (Extended Data Fig. 5a,b). Notably, these newly established COHs were significantly enriched for GO terms related to transmembrane transport and transporter activity, highlighting

293 their role in coordinating adaptive transcriptional programs during stress adaptation
294 (Extended Data Fig. 5c,d).

295 **NapM is essential for chromosome architecture plasticity and** 296 **adaptive remodeling under stress**

297 To understand the plasticity of chromosome architecture and coordinated transcrip-
298 tion reprogramming during stress adaptation, we first compared Hi-C contact maps
299 between wild-type and *napM* knockout strains under the same hypoxic conditions
300 (WT-hypoxia versus *napM*-KO-hypoxia; Fig. 4a–c). Our data showed that *napM*-KO
301 resulted in extensive reorganization of chromosome contacts across the genome. DI
302 profiles and corresponding scalogram representations revealed substantial disruption
303 of higher-order chromosome architecture the *napM*-KO strain under hypoxia (Fig. 4b).
304 As DI captures the preferential upstream or downstream interaction bias of genomic
305 bins, coherent DI patterns were often used to reflect the presence of functional chro-
306 mosome organization units[16, 36]. Our data showed that in the absence of NapM, DI
307 profiles became fragmented and less spatially coherent (Fig. 4b, Fig. 3d), indicating
308 more severe impaired organization of functional chromosome interaction regions. This
309 global disruption was further supported by the Hi-C difference matrix, which showed
310 widespread gain and loss of interactions throughout the chromosome rather than local
311 architectural changes (Fig. 4c).

312 The reconstruction of the chromosome architecture was further applied to explore
313 the structural consequences of the loss of NapM at the whole-genome level (Fig. 4d).
314 While the hypoxia-treated WT strain maintained a relatively ordered configuration
315 compared with the untreated WT, the *napM*-KO strain under hypoxia exhibited pro-
316 found alterations, including asymmetric condensation, irregular folding, and reduced
317 spatial segregation across chromosomal regions. Consistently, transcriptomic analysis
318 revealed substantial reprogramming in the *napM*-KO strain under hypoxia compared
319 with WT-hypoxia (Fig. 4e). Although the *napM*-KO strain remained viable under
320 hypoxic conditions, it failed to survive during the establishment of the latency model
321 (Fig. 4f). To quantitatively evaluate the reorganization of functional chromosome
322 architecture across conditions, we calculated Shannon entropy and one-dimensional
323 Moran's I based on genome-wide DI profiles (Fig. 4g). Shannon entropy measures
324 the global disorder of DI state distributions, whereas Moran's I quantifies the spa-
325 tial autocorrelation of DI values along the chromosome. As shown in Fig. 4g, within
326 the wild-type background, hypoxia led to a moderate increase in Shannon entropy
327 compared to WT (3.49 vs. 3.01), while the latent condition caused a much more
328 pronounced increase (8.97), indicating a substantial rise in chromosome disorder. Con-
329 sistently, the one-dimensional Moran's I decreased from 0.95 (WT) to 0.93 under
330 hypoxia and further dropped to 0.76 in the latent state, reflecting a progressive loss
331 of spatial coherence. In contrast, in the *napM*-KO background, the baseline entropy
332 was already higher than WT (3.68 vs. 3.01). Upon hypoxia, entropy further increased
333 to 5.08, accompanied by a reduction in Moran's I from 0.94 to 0.86, indicating an
334 exacerbation of chromosome disorganization. Notably, the magnitude of change under
335 hypoxia was greater in the KO than in WT, suggesting that loss of *napM* sensi-
336 tizes chromosome architecture to hypoxic stress. Together, these quantitative metrics

337 demonstrate that stress induces moderate remodeling of genome organization in WT,
338 whereas loss of *NapM* markedly exacerbates structural disorder and compels bacte-
339 ria to undergo substantial chromosome reorganization in response to environmental
340 stress.

341 To further investigate how NapM influences higher-order genome compartmental-
342 ization under stress, we performed A/B compartment analysis based on eigenvector
343 decomposition of Hi-C contact matrices (Extended Data Fig. 6c–d). Saddle plots
344 revealed a progressive weakening of A/B compartmental segregation from control
345 to hypoxic and latent conditions, in the wild-type strain, characterized by gradual
346 attenuation of preferential A–A and B–B interactions. Importantly, the chromosome
347 compartment strength also showed a consistent reduction under stress conditions,
348 indicating a programmed decondensation of compartmental organization during stress
349 environmental adaptation (Extended Data Fig. 6e). In contrast, loss of NapM resulted
350 in a dramatically pronounced and aberrant disruption of genome compartmental-
351 ization. The *napM*-KO genome displayed a saddle plot pattern that was obviously
352 distinct from all other conditions, with markedly diminished separation between A and
353 B compartments and extensive mixing of inter-compartment interactions. This effect
354 was also reflected quantitatively by the lowest compartment strength observed across
355 all conditions (bootstrap test, $p < 1 \times 10^{-4}$). Together, these results suggested that
356 WT *M. tuberculosis* may undergo a well-controlled and highly plastic modulation of
357 A/B compartmentalization in response to stress. By contrast, NapM deficiency leads
358 to a collapse of compartmental organization, particularly under hypoxia. This dra-
359 matic aberrant compartmental remodeling further supports a critical role for NapM in
360 maintaining the plasticity of higher-order chromosome architecture and maintaining
361 chromosome reorganization within a reversible range.

362 Across all conditions, WT genomes displayed an ordered increase in entropy accom-
363 panied by a gradual decrease in Moran’s I during hypoxia and latency, reflecting
364 a programmed reorganization of chromosome. In contrast, the *napM*-KO genome
365 exhibited a more uncontrolled range of structural alterations, as evidenced by the
366 pronounced chromosome decondensation observed under hypoxia. Notably, in WT
367 bacteria, adaptation to latent stress requires more extensive chromosome architecture
368 reorganization than that observed under hypoxia. Latent-state entry involves broader
369 transcriptional reprogramming than hypoxia, with 1,922 differentially expressed genes
370 compared to 1,507 under hypoxia, including 896 shared genes (Jaccard index = 0.354)
371 (Fig. 4i,j). Based on this, we hypothesized that the *napM*-KO genome would require
372 an even greater degree of structural remodeling to transition into the latent state,
373 potentially exceeding a tolerable threshold of chromosome organization (Fig. 4h).
374 Such excessive and poorly constrained reorganization may underlie the inability of
375 the *napM*-KO strain to survive under latency, as consistently observed in our stress
376 experiments. To test this hypothesis, we computationally inferred the putative 3D
377 chromosome architecture of the *napM*-KO latent state using a singular value decom-
378 position (SVD)-based framework (Fig. 4k). Specifically, we integrated Hi-C contact
379 matrices from five experimentally observed conditions (WT, WT-hypoxia, WT-latent,
380 *napM*-KO, and *napM*-KO-hypoxia) to model the latent structural component and
381 project the corresponding *napM*-KO latent configuration (see Methods, Section 4.14).

Furthermore, we applied structural superposition to quantify this effect by root mean square deviation (RMSD) between WT and *napM*-KO under different stress conditions. Structural divergence across conditions was quantified by RMSD (Fig. 4l). In WT, RMSD increased from 2.42 under hypoxia to 3.20 in the latent state. Notably, the *napM*-KO strain already exhibited a larger deviation under hypoxia (RMSD = 3.14), exceeding that of WT, while the putative *napM*-KO latent genome displayed the highest divergence (RMSD = 5.80).

Analysis of per-bin chromosome displacement revealed that the latency chromosome transition was associated with significantly increased displacement in the *napM*-KO genome (Fig. 4m,n). Together, these results suggested that *napM* might play a crucial role for maintaining the plasticity of functional chromosome architecture. By constraining chromosome remodeling within a plastic and reversible range, it enables the genome to dynamically transit between distinct and precisely orchestrated structural states, thereby supporting the transcriptional flexibility required for entering and maintaining latency.

3 Summary and Perspective

The sophisticated 3D genome, once considered a hallmark of eukaryotic complexity, is increasingly recognized as a fundamental and ancient principle of chromosome organization. However, the mechanisms underlying its formation, dynamic remodeling, and its role in coordinating transcription in prokaryotes remain poorly understood. This knowledge gap has obscured the evolutionary origins of spatial chromosome regulation. Here, we used *Mycobacterium tuberculosis* as a model prokaryotic system, given its slow growth rate, and delineate a multi-omics interaction network that integrates DNA-DNA, DNA-protein, and protein-protein interactions to resolve the bacterial chromosome. Central to this network is the conserved nucleoid-associated protein NapM, which functions as a key architectural organizer bridging DNA-protein interactions. Loss of NapM leads to widespread disruption of chromosome architecture, desynchronization of long-range operon interactions, and impaired transcriptional coordination, establishing a tight relationship between spatial chromosome structure and transcriptional regulation. Together with the multi-omics interaction network, these findings provide a comprehensive framework for understanding the molecular machinery underlying prokaryotic 3D genome organization and its associated transcriptional regulation.

Importantly, our study reveals that the *M. tuberculosis* chromosome is highly plastic and undergoes stress-responsive remodeling to support transcriptional reprogramming during environmental adaptation. Quantitative analyses using Shannon entropy and Moran's I demonstrate that NapM is essential for maintaining this architectural plasticity. In the absence of NapM, the chromosome becomes rigid in a structurally constrained state characterized by reduced transcriptional flexibility, ultimately impairing dormancy establishment. Further investigation is needed to elucidate the molecular mechanisms by which the dynamic chromosome of *M. tuberculosis* coordinates transcriptional plasticity, as well as to explore the emergence and evolution of chromosome organization across prokaryotes and eukaryotes. These findings provide

425 a foundation for comparative 3D genomics across the tree of life and may inform the
426 rational design of minimal genomes and synthetic regulatory circuits.

427 Collectively, these findings provide new insights into the principles governing
428 prokaryotic chromosome organization and its functional integration with transcrip-
429 tional regulation. We propose that NapM represents an evolutionarily conserved class
430 of chromosome architectural factors that coordinate transcription through spatial
431 chromosome organization. By revealing how a plastic chromosome facilitates adaptive
432 transcriptional responses, our work offers a conceptual framework for understanding
433 the evolutionary emergence of genome architecture-based gene regulation and its role
434 in bacterial physiology and persistence.

435 4 Methods

436 4.1 Construction of *M. tuberculosis* Latency Culture Model

437 *M. tuberculosis* H37Ra was cultured in Middlebrook 7H9 broth (Becton, Dickinson)
438 supplemented with 0.5% glycerol, 0.05% Tween 80, and 10% oleic acid-albumin-
439 dextrose-catalase (OADC; Becton, Dickinson). Cultivation started with an initial
440 optical density (OD₆₀₀) of 0.1 in 10 mL 7H9 medium in culture flasks, briefly shaken
441 daily to ensure sufficient oxygenation until an OD of 1.0 was reached. Bacteria were
442 pelleted (3,000 × g, room temperature [RT], 10 min), resuspended in 5 mL phosphate-
443 buffered saline (PBS, pH 5.3–5.5), then transferred into 10 mL fresh 7H9 medium (pH
444 5.3–5.5) in a 15 mL tube and incubated for 24 hours. Subsequently, an antibiotic cock-
445 tail ($\frac{1}{2}$ MIC rifampicin, $\frac{1}{2}$ MIC isoniazid, $\frac{1}{3}$ MIC kanamycin, $\frac{1}{4}$ MIC streptomycin) was
446 added, and incubation continued for two weeks. Latency induction was confirmed by
447 inoculating 0.1 mL culture into fresh 7H9 broth (5 mL), monitoring OD every second
448 day for six days without growth. Viability was validated using the Alamar Blue assay.

449 4.2 Construction of *M. tuberculosis* Hypoxia Culture Model

450 Cultivation began with an initial OD₆₀₀ of 0.1 in 10 mL 7H9 medium, with daily brief
451 shaking until reaching OD₆₀₀ of 0.8. Bacteria were then transferred to a 15 mL tube,
452 completely filled with fresh 7H9 medium to eliminate headspace, tightly sealed, and
453 incubated for 13 days. Hypoxic conditions were confirmed by complete decolorization
454 of methylene blue (1.5 µg/mL), which typically occurred after five days of incubation.
455 Following this confirmation, the bacteria were maintained under hypoxic conditions
456 for an additional seven days.

457 4.3 RNA Isolation and RNA-seq Library Preparation

458 Total RNA extraction was performed using RNAiso Plus (Takara, Cat. 9109), fol-
459 lowing the manufacturer’s instructions. Ribosomal RNA was depleted with Ribo-Zero
460 rRNA Removal Reagent (Bacteria; Illumina, Cat. RZMB12324). RNA-seq libraries
461 were generated with the VAHTS Universal V8 RNA-seq Library Prep Kit (Vazyme,
462 Cat. NRM605), adhering to the manufacturer’s protocol.

463 4.4 Bacterial Hi-C Assay

464 **Preparation and fixation of bacterial *M. tuberculosis* cultures.** Exponential-
465 phase H37Ra cultures (10 mL) were transferred to pre-chilled tubes, pelleted by
466 centrifugation (3,500 rpm, 4°C, 5 min), washed with pre-cooled PBS, and fixed with
467 3% formaldehyde (Sigma, Cat. F8775) for 1 hour at 4°C on a roller bench (30 rpm).
468 Glycine (final 0.375 M) was then added to quench fixation for 15 min at 4°C, and
469 bacteria were then pelleted again.

470 **Preparation of cell lysates.** Pellets were resuspended in 45 μ L 1 $\times$ TE buffer,
471 supplemented with 5 μ L lysozyme, and incubated overnight (37°C, 1000 rpm). Next,
472 10 μ L of 10% SDS was added, followed by incubation at 55°C for 1 hour (1000 rpm).

473 **DpnII digestion.** Digestion mixture (25 μ L 10 $\times$ DpnII buffer, 25 μ L 10% Triton
474 X-100, 136 μ L nuclease-free water) was added to the lysate (53 μ L), incubated (37°C,
475 10 min, 1000 rpm), followed by addition of 25 μ L DpnII (NEB) enzyme and incubation
476 for 4 hours at 37°C (450 rpm).

477 **Filling restriction ends.** The digested DNA ends were filled using 7 μ L dNTPs
478 (10 mM each), 1.5 μ L BSA (20 mg/mL), 43 μ L nuclease-free water, and 6 μ L Klenow
479 DNA Polymerase (Vazyme), incubating at 25°C for 45 min (450 rpm). Reaction was
480 stopped with 15.8 μ L 10% SDS at 25°C for 20 min (1000 rpm).

481 **Proximity ligation.** chromosome was fractionated by centrifugation (16,000 rpm,
482 1 h, 4°C), pellet resuspended, DNA quantified by Qubit dsDNA HS Assay. For ligation,
483 2 μ g chromosome in 820 μ L nuclease-free water was mixed with 100 μ L 10 $\times$ T4 ligase
484 buffer, 75 μ L 10% Triton X-100, 5 μ L BSA (20 mg/mL), incubated at 16°C for 20
485 min (1000 rpm), followed by ligation with 5 μ L T4 DNA Ligase (TAKARA) overnight
486 at 16°C. Subsequently, 16.6 μ L RNase A (10 mg/mL) and 12.5 μ L Proteinase K (20
487 mg/mL) with 120.8 μ L NaCl (5 M) were added, incubating at 65°C for 6 hours. DNA
488 was extracted using phenol-chloroform, quantified by Qubit.

489 **Hi-C library construction.** DNA fragmentation (300–500 bp) was performed
490 using Bioruptor Pico and purified with VAHTS DNA Clean Beads (Vazyme, Cat.
491 N411-01). DNA fragments were end-repaired and adenylated using End Prep Mix
492 (Vazyme), ligated to Illumina sequencing adapters with T4 DNA Ligase in PEG4000-
493 containing buffer for 2 hours at room temperature, and purified. Libraries were
494 amplified (10 cycles), purified again with VAHTS beads, and sequenced by Illumina
495 paired-end sequencing.

496 4.5 Hi-C Data Processing and chromosome Reconstruction

497 Raw Hi-C sequencing data (FASTQ format) were processed using **HiC-Pro** (v3.1.0)
498 to map paired-end reads to the reference genome, filter valid interaction pairs, per-
499 form quality control, and generate genome-wide intra- and inter-chromosomal contact
500 matrices. The processed outputs were subsequently converted to .mcool and .hic
501 formats at resolutions of 1 kb, 5 kb, and 10 kb using **cooler** (v0.8.2) [37] and **Juicer**
502 **Tools** (v1.22.01)[38], respectively. Visualization of contact matrices and differential
503 interactions was conducted using **CoolBox** (v0.3.9)[39].

504 The chromosome models of *Mycobacterium tuberculosis* chromosome were recon-
505 structed using the **LorDG**[40] algorithm, and visualization of the resulting structures
506 was performed using **PyMOL** (v3.1.3)[41].

507 Pearson correlation analysis of contact matrices across biological replicates was per-
508 formed using the **hicCorrelate** function implemented in **HiCExplorer** (v3.7.2)[42],
509 with visualization achieved via scatter plots.

510 Loops were identified by the **hiccup**s algorithm within **Juicer Tools**. CIDs were
511 defined based on the directionality index (DI) score method described by Dixon et
512 al. (2012), using 100 kb binning to infer directional biases consistent with CID bound-
513 aries. Domain boundaries and interaction domains were subsequently delineated by
514 applying a Hidden Markov Model (HMM) to identify hidden directionality biases
515 (DB), with domains and boundaries inferred from genome-wide HMM state calls[36].

516 4.6 Gene expression coordination analysis within CIDs and 517 across loops

518 **Gene expression variance within CIDs.** To assess whether transcription is coor-
519 dinated within CIDs, we quantified expression variability across 5-kb genomic bins
520 inside each CID and compared it to the matched random control. Genome-wide expres-
521 sion was quantified from RNA-seq as normalized expression values assigned to 5-kb
522 bins. For each CID, all bins fully contained within the CID span were extracted, and
523 the variance of bin-level gene expression values was computed. The mean expression
524 variance across all CIDs was then calculated as the observed CID-level variability.

525 To generate a random expectation while preserving CID size distribution, we
526 performed random bin resampling. Specifically, conducted a sampling without replace-
527 ment to obtain a pool of bins equal in total number to those in all CIDs. Then, we
528 partitioned this sampled pool into groups corresponding to the empirical bin counts of
529 each CID. This procedure was repeated 50 times, yielding a distribution of mean gene
530 expression variances expected under random grouping. Statistical significance between
531 observed CID variances and randomized variances was assessed using a two-sided
532 Welch's *t*-test (unequal variance).

533 **Co-expression analysis of loop anchor genes.** To determine whether loops
534 connect transcriptionally coordinated loci, we compared gene expression similarity
535 between loop anchor genes and the distance-matched random controls. For each loop,
536 the two anchors (A and B) were mapped to the corresponding 5-kb expression bins,
537 and expression similarity was quantified as the ratio of their expression levels:

$$\text{Ratio}_{AB} = \frac{\max(\text{exp}_A, \text{exp}_B)}{\min(\text{exp}_A, \text{exp}_B)} - 1.$$

538 To generate a matched null distribution, we fixed one anchor (A) and randomly
539 selected a control locus (C). Expression similarity for each random pair was com-
540 puted as Ratio_{AC} using the same formula. This randomization was repeated across 10
541 independent iterations to obtain a robust null distribution.

542 Differences between the observed loop-anchor ratios (Ratio_{AB}) and the random
543 distance-matched ratios (Ratio_{AC}) were evaluated using a two-sided Welch's *t*-test.

Boxplots were generated using `seaborn`, and statistical significance was reported as exact P values.

4.7 Operon Interaction Analysis

A total of 749 operons were identified according to previously reported annotations[33]. Operon-to-operon interaction strengths were calculated by integrating operon length-based weighting factors with the genome-wide 3D chromosome interaction matrices. Specifically, interaction strengths were weighted proportionally to operon length, resulting in a comprehensive operon interaction matrix. Given that the chromosome of *Mycobacterium tuberculosis* exhibits a U-shaped conformation, operons located within 5% of genome length at both genome termini were excluded from the analysis. Furthermore, interactions were retained only for operon pairs separated by genomic distances greater than 1% of the total genome length.

To identify high-confidence operon interactions, a statistical threshold was established based on the distribution of off-diagonal interaction strengths from the interaction matrix. Specifically, the threshold was defined as $\mu + 2\sigma$, where μ represents the mean, and σ the standard deviation of these interaction values:

$$\text{Threshold}_{\text{interaction}} = \mu + 2\sigma \quad (1)$$

Subsequently, operon pairs exhibiting interaction strengths above this threshold were further filtered based on expression similarity. Expression similarity between operon pairs was calculated as follows:

$$\text{Similarity} = \frac{\min(\text{exp}_1, \text{exp}_2)}{\max(\text{exp}_1, \text{exp}_2)} \quad (2)$$

Operon pairs with expression similarity greater than 0.9 were retained as final candidate interacting pairs for downstream analyses.

4.8 Homology analysis of Rv0047c

All homologous gene data of Rv0047c analyzed in this study were obtained from the OrthoDB database, comprising a total of 2,455 genes across 2,348 bacterial species. Sequence similarity searches were performed using the local BLAST tool provided by NCBI. Visualization of the results was carried out in Python using the Matplotlib package.

4.9 Transcriptome Data Analysis

RNA-seq data were processed using the `nf-core` RNA-seq analysis pipeline implemented via Nextflow (version 24.10.2.5932)[43]. Sequencing reads were aligned to the reference genome using `HISAT2`, and gene-level counts were generated using `featureCounts`. Differential gene expression analysis was subsequently performed using the `limma` package (version 3.36.2)[44] in R(4.3.2). Genes exhibiting an adjusted p -value (q -value) < 0.05 and an absolute $\log_2$ fold-change greater than 1 were identified as significantly differentially expressed.

579 4.10 Integration of protein–protein and protein–DNA 580 interaction datasets

581 To construct a comprehensive multi-layered interaction network, we integrated
582 three experimentally validated datasets including the protein–protein (PPI) and
583 protein–DNA (PD) interactomes of *Mycobacterium tuberculosis*.

584 The global PPI network was derived from previous study[34], which identified
585 more than 8,000 high-confidence protein–protein interactions connecting 2,907 pro-
586 teins—representing approximately 74% of the functional proteome—using large-scale
587 yeast two-hybrid screening combined with biochemical validation.

588 The transcriptional regulatory (TR) network was obtained from Zeng *et al.* (2012)
589 [33], which mapped more than 5,400 transcription factor (TF)–promoter interactions
590 covering nearly all TFs and one-third of *M. tuberculosis* genes. This dataset provided
591 a comprehensive reference framework for global regulatory network reconstruction.

592 Additionally, the genome-wide DNA-binding network reported by Minch
593 *et al.* (2015) [35] was incorporated, encompassing ~16,000 high-confidence pro-
594 tein–DNA interactions from 154 DNA-binding proteins profiled by ChIP–seq. This
595 study highlighted abundant DNA-binding proteins (each recognizing >500 sites) and
596 demonstrated the extensive chromosomal occupancy of transcriptional regulators in
597 *M. tuberculosis*.

598 Integration was performed with **networkx** (v3.1) [45] to generate a unified network
599 containing 2,992 proteins, 8,043 PPIs, and 17,223 PD interactions.

600 4.11 Quantification of interaction-change amplitude between 601 DEG and stable gene pairs

602 DEGs were defined from the comparison between the condition of interest and wild
603 type (WT) using the gene-level differential expression table. Genes were classified as
604 DEGs if they satisfied

$$|\log_2(\text{FC})| \geq \theta_{\text{DEG}} \quad \text{and} \quad q \leq q_0,$$

605 where $\theta_{\text{DEG}} = 1.0$ and $q_0 = 0.01$. Genes with stable expression were defined as

$$|\log_2(\text{FC})| \leq \theta_{\text{stable}} \quad \text{and} \quad q > q_0,$$

606 where $\theta_{\text{stable}} = 1.0$. The DEG set and the stable-gene set were enforced to be mutually
607 exclusive.

608 Gene genomic coordinates were extracted from the **GTF** annotation file. Each gene
609 was assigned to a 5-kb genomic bin according to the midpoint of the annotated gene
610 body:

$$m_g = \left\lfloor \frac{s_g + e_g}{2} \right\rfloor,$$

611 where s_g and e_g denote the start and end coordinates of gene g , respectively. A gene
612 was assigned to the bin whose interval contained m_g .

613 To quantify pairwise chromosome interaction changes, all possible within-group
614 gene pairs were generated separately for the DEG set and the stable-gene set, yielding
615 DEG–DEG pairs and Stable–Stable pairs. If two genes fell into the same 5-kb bin, that
616 pair was excluded because bin-level interaction could not resolve an intra-bin interac-
617 tion. When multiple gene pairs mapped to the same bin pair, they were collapsed into
618 a single bin–bin interaction to avoid repeated counting of the same Hi-C observation.
619 For each bin pair (i, j) , contact intensities were extracted from the WT and
620 condition-specific cooler matrices, denoted as C_{ij}^{WT} and C_{ij}^{Cond} , respectively. Low-
621 signal interactions were removed using a shared threshold estimated jointly from both
622 DEG–DEG and Stable–Stable bin pairs. Specifically, for each pair we considered

$$M_{ij} = \max(C_{ij}^{\text{WT}}, C_{ij}^{\text{Cond}}),$$

623 and retained only pairs with

$$M_{ij} \geq T,$$

624 where T was defined as the 45th percentile of the pooled M_{ij} values from both groups.
625 Because chromosome contact frequency strongly depends on genomic distance,
626 DEG–DEG and Stable–Stable pairs were distance-matched before statistical com-
627 parison. For each genomic separation category, defined as the difference in bin
628 indices

$$d_{ij} = j - i,$$

629 the same number of DEG–DEG and Stable–Stable pairs was randomly sampled
630 without replacement, using the smaller group size within that distance class. This pro-
631 cedure ensured that the genomic distance distributions were comparable between the
632 two groups.

633 For each retained bin pair, the magnitude of interaction change between the
634 condition of interest and WT was quantified as the absolute $\log_2$ fold change:

$$A_{ij} = \left| \log_2 \left(\frac{C_{ij}^{\text{Cond}} + \epsilon}{C_{ij}^{\text{WT}} + \epsilon} \right) \right|,$$

635 where $\epsilon = 0.1$ is a pseudocount added to avoid division by zero. Thus, larger A_{ij} values
636 indicate stronger interaction changes irrespective of direction.

637 To test the hypothesis that interactions among DEGs exhibit larger changes than
638 those among stable genes, the distributions of A_{ij} were compared between the DEG–
639 DEG and Stable–Stable groups using a one-sided Mann–Whitney U test with the
640 alternative hypothesis

$$A_{\text{DEG–DEG}} > A_{\text{Stable–Stable}}.$$

641 For completeness, the two-sided Mann–Whitney U test was also recorded. Effect size
642 was quantified using Cliff’s delta. Results were visualized using violin plots overlaid
643 with jittered individual observations, and the corresponding one-sided p value was
644 annotated on the plot.

645 All analyses were performed in Python using `pandas`, `NumPy`, `SciPy`, `cooler`,
646 `matplotlib`, and `seaborn`.

647 4.12 A/B compartment analysis

648 A/B compartment analysis of all Hi-C datasets was performed using the `cooler` and
649 `cooltools` software packages [37, 46]. Balanced contact matrices at 5-kb resolution
650 were used for all samples. Only cis interactions were considered; the first two diago-
651 nals were excluded, and the maximum genomic separation was limited to 500 kb to
652 minimize local proximity effects.

653 A/B compartments were identified by eigenvector decomposition of the observed/-
654 expected (O/E) cis contact matrices. Specifically, the first eigenvector (E1) of the
655 cis interaction matrix was computed for each chromosome. The sign of E1 was ori-
656 ented using genomic GC content such that positive values corresponded to the A
657 compartment [47].

658 Genomic bins were ranked by their E1 values and divided into 20 quantile
659 bins spanning the range from 0.02 to 0.98. Saddle plots were generated using
660 `cooltools.saddle`. Contact frequencies were first normalized by the smoothed cis-
661 expected values and subsequently scaled by the median of the saddle matrix, resulting
662 in a background interaction level close to 1.

663 From the normalized saddle matrices, interaction pixels corresponding to A–A, B–
664 B, and A–B compartment pairs were extracted. A and B compartments were defined
665 as the highest and lowest 20% of E1 quantiles, respectively. Compartment strength
666 was quantified as:

$$CS = \log_2 \left(\frac{\mu_{AA} \cdot \mu_{BB}}{(\mu_{AB} + \epsilon)^2} \right),$$

667 where μ_{AA} , μ_{BB} , and μ_{AB} denote the mean normalized contact frequencies between
668 A–A, B–B, and A–B compartment pairs, respectively. A small pseudocount ($\epsilon = 10^{-6}$)
669 was added to avoid numerical instability due to zero value[48].

670 Uncertainty in compartment strength estimates was assessed using non-parametric
671 bootstrap resampling. Interaction pixels corresponding to A–A, B–B, and A–B cate-
672 gories were resampled with replacement 10,000 times, and compartment strength was
673 recalculated for each bootstrap iteration. Final estimates were reported as the mean
674 of the bootstrap distribution together with the 95% confidence interval (2.5th–97.5th
675 percentiles).

676 Differences in compartment strength between various conditions were evaluated
677 using a two-sided bootstrap test by assessing whether the bootstrap distribution of the
678 difference between conditions crossed zero. To avoid zero-valued P estimates arising
679 from finite resampling, a continuity correction was applied. Given the total of 10,000
680 bootstrap iterations, the minimum resolvable empirical P value was limited, and values
681 smaller than 1×10^{-4} are uniformly reported as $P < 1 \times 10^{-4}$ in figures.

682 4.13 Calculation of Shannon Entropy and 1D Moran's I

683 To quantify the degree of structural order within CIDs, Shannon entropy and the one-
684 dimensional Moran's I were calculated based on the Directionality Index (DI) scores.
685 Specifically, Shannon entropy was applied to measure the level of orderliness in DI
686 distributions, while Moran's I assessed the spatial autocorrelation of DI scores across
687 genomic bins. Higher structural organization, indicated by clearer CID boundaries,

688 is characterized by lower Shannon entropy (greater orderliness) and higher Moran's I
689 values (stronger spatial coherence).
690 Shannon entropy was calculated as follows:

$$H = - \sum_i p_i \log_2(p_i) \quad (3)$$

691 where p_i represents the frequency of discretized DI scores.
692 The one-dimensional Moran's I was computed to measure spatial autocorrelation
693 among adjacent genomic bins:

$$I = \frac{n}{\sum_{i,j} w_{ij}} \frac{\sum_{i,j} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \quad (4)$$

694 where x_i denotes the DI score at genomic bin i , $\bar{x}$ is the mean DI score across bins,
695 n is the total number of bins, and w_{ij} represents the adjacency-based weighting (with
696 a value of 1 if bins i and j are neighbors, otherwise 0).

697 4.14 Prediction of the counterfactual *napM*-KO Latent 698 contact matrix

699 To infer the putative chromosome contact landscape of *Mycobacterium tuberculosis* in
700 a *napM*-KO latent state, we employed a counterfactual modeling framework based on
701 singular value decomposition (SVD). Hi-C contact matrices were generated at 5-kb
702 resolution for five experimental conditions (WT, WT-Hypoxia, WT-Latent, *napM*-
703 KO, and *napM*-KO-Hypoxia). Each matrix was log-transformed using $\log(1 + \text{counts})$
704 and vectorized by extracting the upper-triangular entries. A design matrix was then
705 constructed to encode wild-type versus *napM*-KO status as well as hypoxic versus
706 latent stress components.

707 Model training was performed using a leave-one-condition-out cross-validation
708 strategy to select the optimal latent rank. The SVD-based imputation achieved the
709 lowest average root-mean-square error (RMSE) at rank 1, indicating that a single
710 dominant latent factor captured the majority of condition-level variability. Using
711 the fitted model, we projected the latent coordinate corresponding to the *napM*-
712 KO latent design and reconstructed the predicted contact matrix, followed by inverse
713 transformation (`expm1`) and rounding to integer counts.

714 4.15 Kabsch alignment and RMSD calculation

715 To quantitatively assess conformational changes between conditions, we performed
716 structural superposition using the Kabsch algorithm. For each pairwise comparison
717 (e.g., WT vs. Hypoxia or WT vs. Latent), the two 3D coordinate sets were first cen-
718 tered by subtracting their centroids. The optimal rotation matrix was then derived by
719 singular value decomposition (SVD) of the covariance matrix, ensuring a right-handed
720 coordinate system by correcting the determinant. Applying this rotation followed by
721 translation aligned the query structure onto the reference (WT).

722 After alignment, the global root-mean-square deviation (RMSD) was computed as:

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\mathbf{r}_i^{\text{ref}} - \mathbf{r}_i^{\text{aln}}\|^2},$$

where N is the number of atoms, $\mathbf{r}_i^{\text{ref}}$ denotes the reference coordinates (WT), and $\mathbf{r}_i^{\text{aln}}$ the aligned coordinates. This value provides a global measure of conformational deviation.

In addition, we computed per-atom displacement vectors after alignment, defined as:

$$d_i = \|\mathbf{r}_i^{\text{ref}} - \mathbf{r}_i^{\text{aln}}\|,$$

which quantify the local positional differences for each atom. These per-atom deviations were further summarized as distributions to compare the magnitude of structural shifts under different conditions.

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Declarations

Conflict of interest / Competing interests The authors declare no competing interests.

Ethics approval and consent to participate Not applicable.

Data availability All raw sequencing data have been deposited in the NCBI Sequence Read Archive (SRA) under accession number PRJNA1452606. Hi-C intermediate processing files and analysis scripts are available at https://github.com/wikk-chy/mtb_hic

Materials availability All materials and strains used in this study are available from the corresponding author upon reasonable request.

Code availability All analysis scripts and custom code used in this study are available at https://github.com/wikk-chy/mtb_hic.

Author contributions

G.C., A.R., and H.C. conceived the study and designed the overall research framework. A.R., Y.W., X.C., and X.C. performed the experiments, including sample preparation, sequencing library construction, and validation assays. H.C., P.W., and

758 Y.C. carried out data processing, multi-omics integration, and computational analyses,
759 including chromosome reconstruction, network modeling, and statistical quantifica-
760 tion. They also performed data visualization and figure generation. H.C., G.C., A.R.,
761 and P.W. interpreted the results, assembled the figures, and wrote the manuscript,
762 incorporating feedback from all authors. W.X., Y.Z., and Y.X. contributed to data
763 interpretation and critically revised the manuscript. G.C. and A.R. supervised the
764 study and provided overall guidance throughout the project.

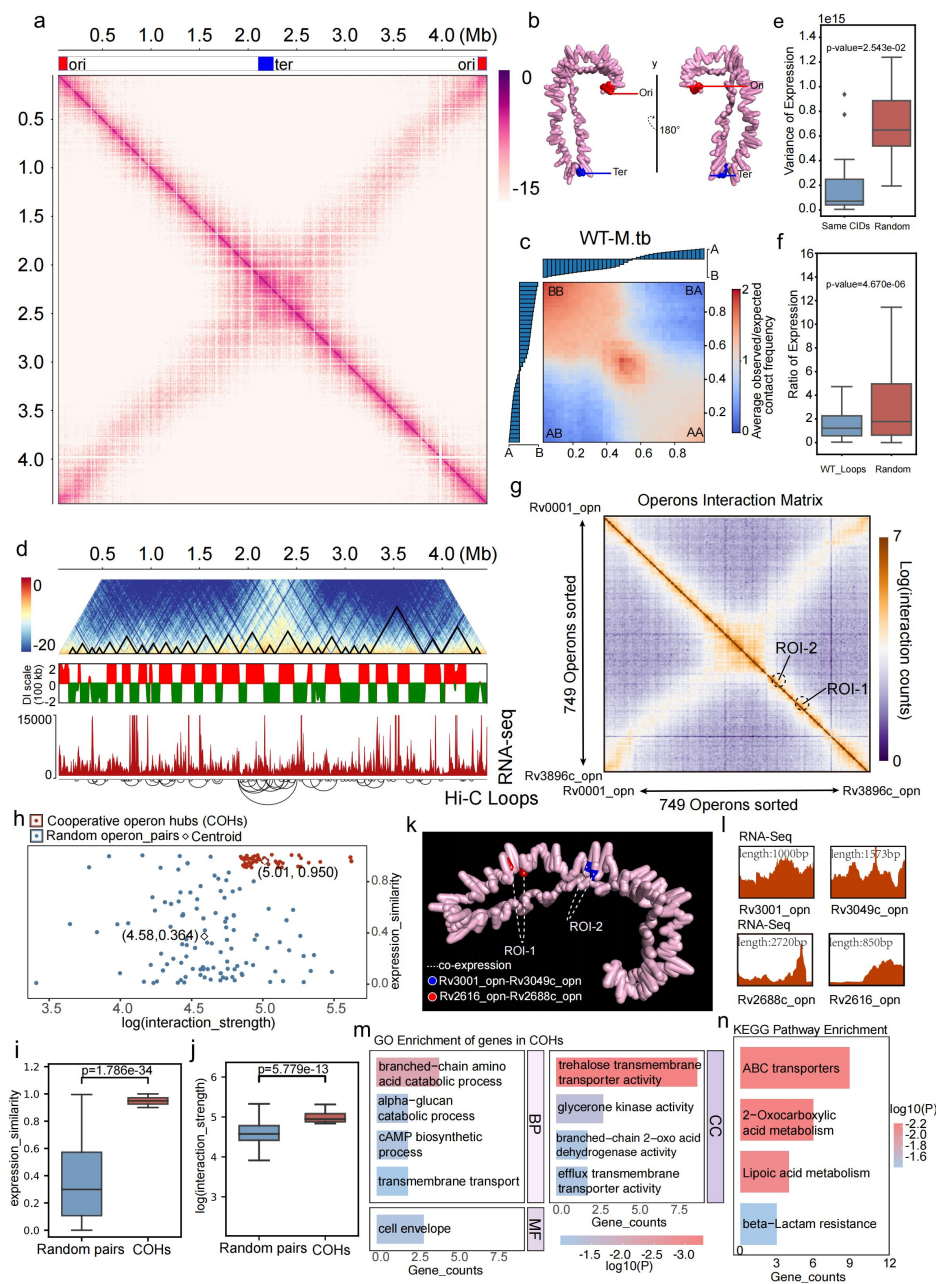


Fig.1 Chromosome architecture and its coordinated transcription in *Mycobacterium tuberculosis*.

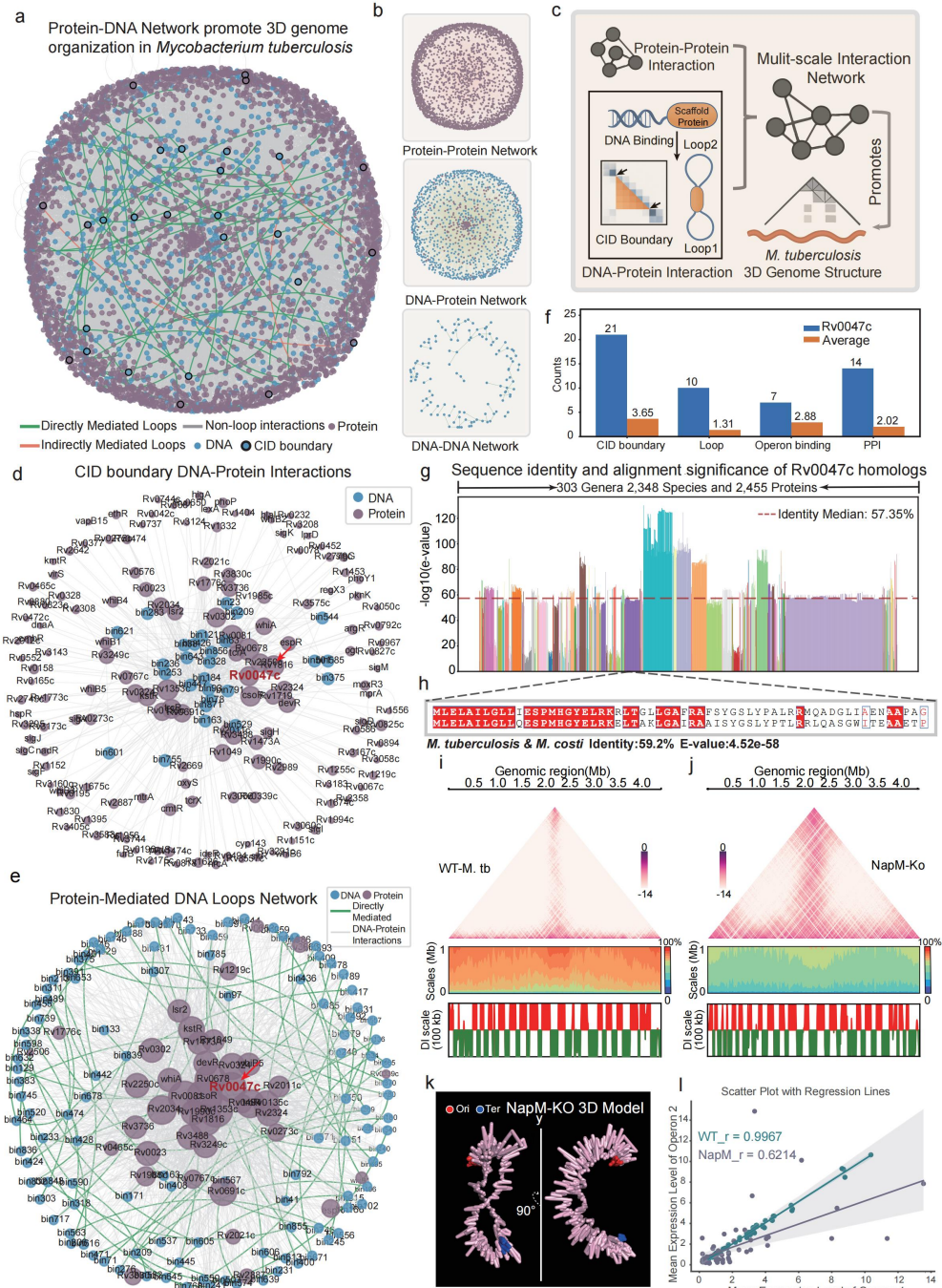


Fig.2 Integrated protein-protein, protein-DNA, and DNA-DNA multi-layer interaction network underlying 3D genome establishment in *Mycobacterium tuberculosis*.

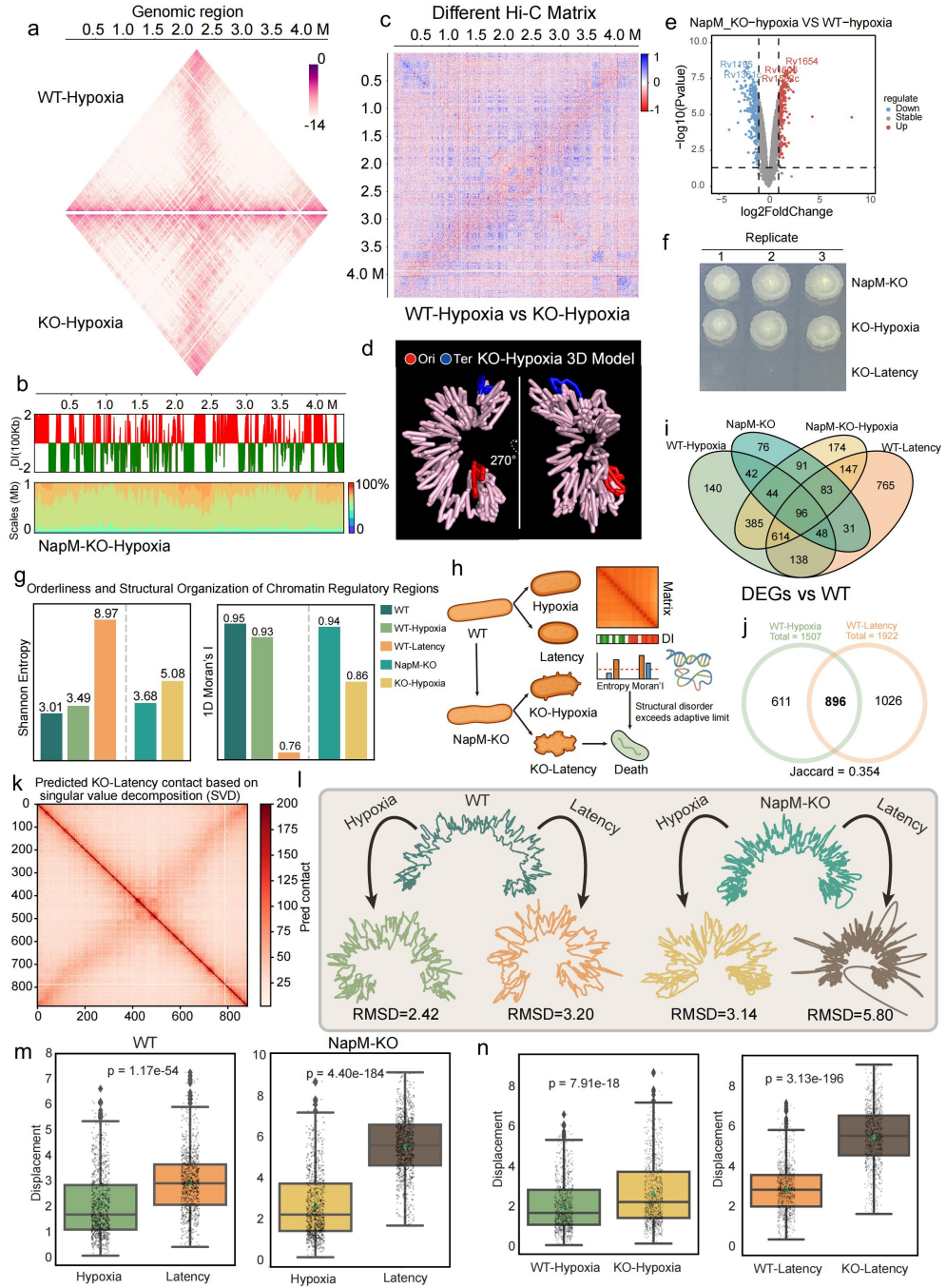


Fig.4 NapM is required for chromosome architecture plasticity and coordinated transcriptional reprogramming.

766 Figure Legends

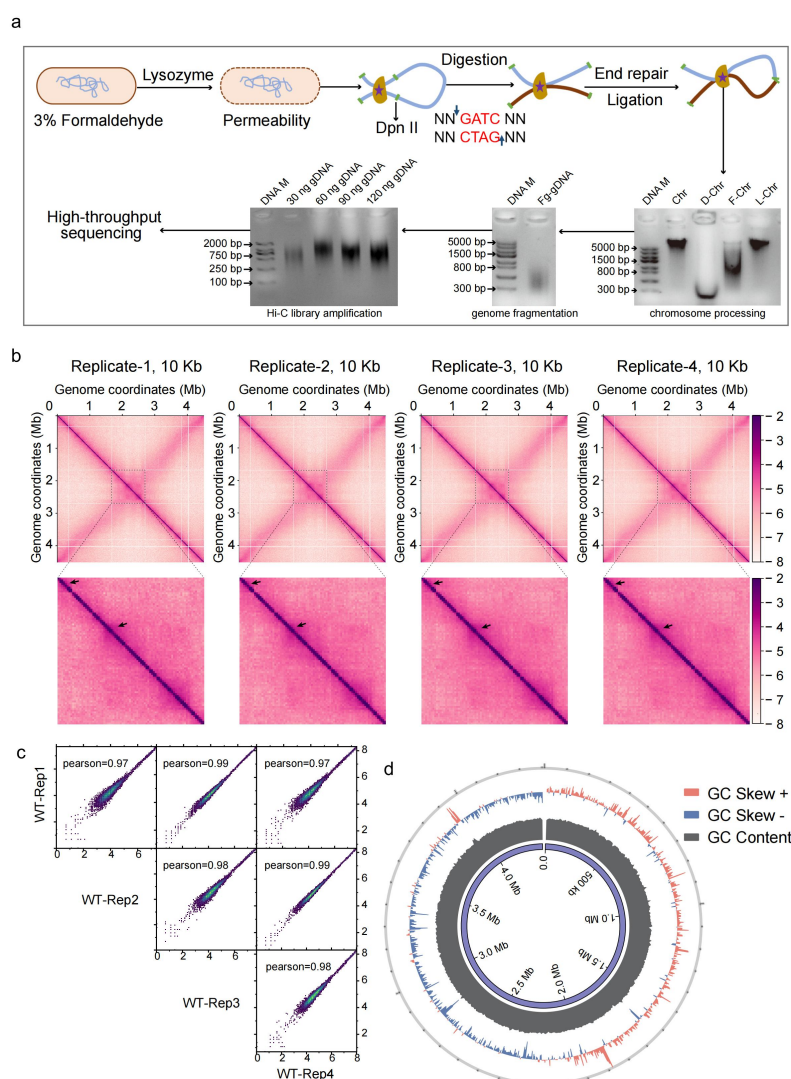
767 **Figure 1. Chromosome architecture and its coordinated transcription in**
 768 *Mycobacterium tuberculosis*. **a,b**, Genome-wide Hi-C contact map and recon-
 769 structed 3D structure of *M. tuberculosis*, revealing a longitudinally folded chromosome
 770 with a characteristic ori-ter organization. **c**, Saddle plot analysis showing preferential
 771 A-A and B-B interactions, indicative of higher-order A/B compartmentalization. **d**,
 772 Integrated genome-wide profiles of CIDs, directionality index (DI), RNA-seq expres-
 773 sion, and loops. **e,f**, Genes within the same CIDs and loop anchors exhibit higher
 774 co-expression relative to randomized controls. **g**, Operon-operon interaction heatmap
 775 derived from Hi-C data for 749 annotated operons, with two highlighted interac-
 776 tion sites. **h-j**, Identification of COHs, showing significantly enhanced interaction
 777 strength and expression synchrony compared with random operon pairs. **k,l**, chro-
 778 mosome reconstruction and operon-level RNA-seq profiles of representative COHs
 779 (*Rv3001_opn-Rv3049c_opn* and *Rv2616_opn-Rv2688c_opn*), illustrating spatial prox-
 780 imity and coordinated gene transcription. **m,n**, Gene Ontology and KEGG pathway
 781 enrichment analyses of COH-associated genes, revealing enrichment in transport- and
 782 metabolism-related functional categories.

783 **Figure 2. Integrated protein-protein, protein-DNA, and DNA-DNA**
 784 **multi-layer interaction network underlying 3D genome establishment in**
 785 *Mycobacterium tuberculosis*. **a**, Integrated interaction network combining protein-
 786 protein, protein-DNA, and DNA-DNA interactions; CID boundaries and loop anchors
 787 are annotated with their associated binding proteins. **b**, Three subnetworks extracted
 788 from panel a: DNA-DNA, protein-protein, and DNA-protein. **c**, Schematic illustrat-
 789 ing how coupled DNA and protein interactions organize the bacterial chromosome. **d**,
 790 Interaction network of proteins bound at CID boundaries. **e**, Interaction network of
 791 proteins bound at loop anchors. **f**, Enrichment of Rv0047c (NapM) binding at CID
 792 boundaries, loop anchors, associated operons and interacting proteins; bars compare
 793 NapM counts to the proteome-wide average. **g**, Conservation analysis of Rv0047c
 794 across bacteria. **h**, Amino-acid sequence conservation of Rv0047c between *M. tubercu-*
 795 *losis* and *M. costi*. **i,j**, Hi-C contact maps, cumulative contact scores, and directionality
 796 index (DI) profiles comparing WT and *napM*-KO. **k**, 3D structural model simulated
 797 for the *napM*-KO. **l**, Loss of long-range operon gene co-expression in the *napM*-KO,
 798 indicating reduced cooperative operon interactions.

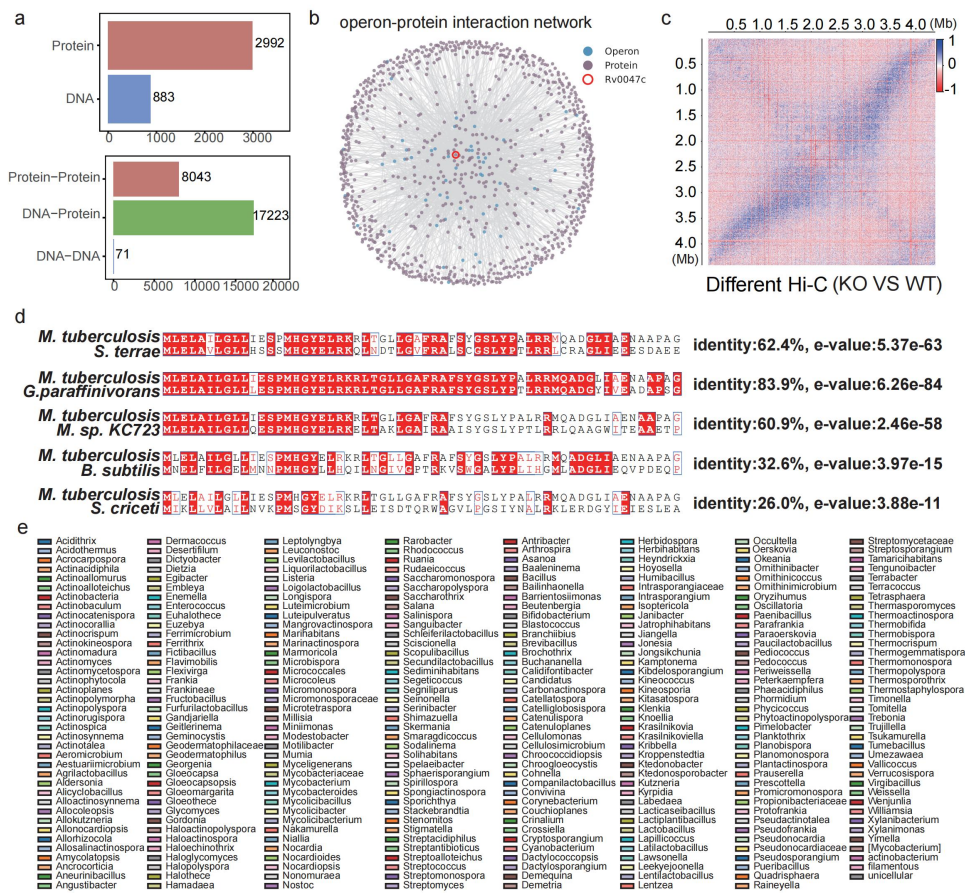
799 **Figure 3. Coordinated reprogramming of chromosome architecture and**
 800 **gene transcription in response to stress conditions.** **a,b**, Genome-wide Hi-C
 801 contact maps and corresponding difference matrices comparing wild-type (WT) cells
 802 with WT under hypoxic (**a**) and latent (**b**) conditions. **c**, Cumulative contact proba-
 803 bility decay curves for WT, WT-hypoxia, and WT-latent, reflecting stress-dependent
 804 alterations in chromosome folding strength. **d**, DI profiles showing remodeling of CIDs
 805 under hypoxia and latency conditions. **e**, chromosome reconstructions under hypoxic
 806 and latent conditions. **f**, Quantification of ori-ter interaction strength across differ-
 807 ent stress conditions, where increased contacts indicate longitudinal compaction and
 808 decreased contacts reflect chromosome decondensation. **g,h**, Volcano plots of DEGs

809 for WT-hypoxia versus WT (**g**) and WT-latent versus WT (**h**). **i,j**, Comparison of
810 chromosome interaction change amplitudes between DEG–DEG and Stable–Stable
811 gene pairs. The y-axis represents the absolute $\log_2$ fold change of contact intensity
812 between the condition and WT, and significance was assessed using a one-sided Mann–
813 Whitney U test. **k**, Under hypoxia, transcriptional coordination between operon pairs
814 within WT COHs is disrupted. **l**, Under latent conditions, transcriptional coordination
815 between operon pairs within WT COHs is disrupted.

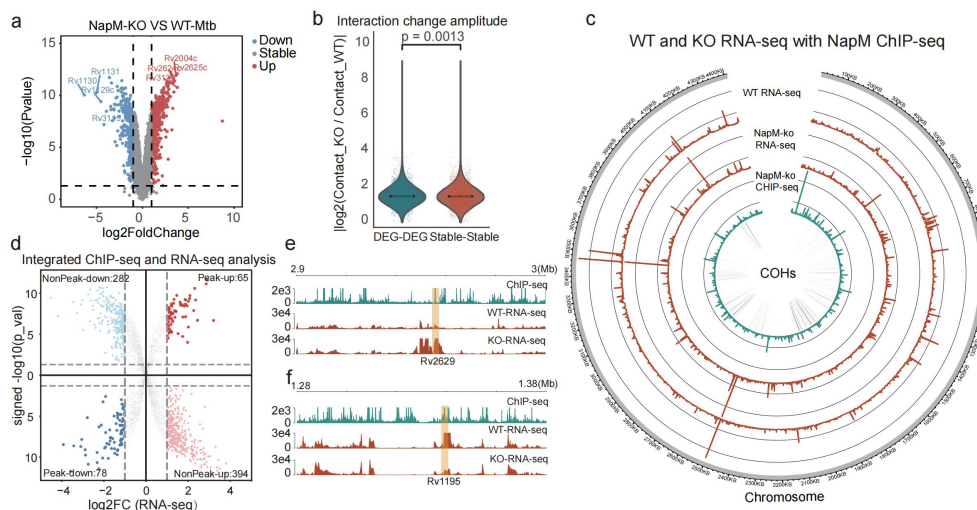
816 **Figure 4. NapM is required for chromosome architecture plasticity and**
817 **coordinated transcriptional reprogramming.** **a–c**, Genome-wide Hi-C contact
818 maps, directionality index (DI) profiles, and difference matrix comparing WT-hypoxia
819 and *napM*-KO-hypoxia, revealing widespread disruption of chromosome organization
820 upon NapM loss. **d**, chromosome structure of the *napM*-KO strain under hypoxic con-
821 ditions, reconstructed from Hi-C data and visualized to illustrate global chromosome
822 organization. **e**, Volcano plot of DEGs between *napM*-KO-hypoxia and WT-hypoxia.
823 **f**, Spot assay showing that *napM*-KO strains remain viable under hypoxia but fail to
824 survive under prolonged latent conditions. **g**, Quantification of chromosome order and
825 plasticity using Shannon entropy and Moran’s I across different conditions, indicat-
826 ing constrained adaptive remodeling plasticity in the absence of NapM. **h**, Conceptual
827 model integrating Shannon entropy and Moran’s I measurements, illustrating that
828 the extent of chromosome reorganization required for the *napM*-KO strain to enter
829 the latent state would exceed the permissible range of chromosome plasticity, thereby
830 resulting in loss of viability. **i,j**, Overlap of DEGs across stress conditions, showing
831 that latent-state entry requires broader transcriptional reprogramming than hypoxic
832 adaptation. **k**, Counterfactual Hi-C contact matrix of the putative *napM*-KO latent
833 state inferred by singular value decomposition (SVD). **l–n**, Quantification of chromo-
834 some structural divergence by RMSD and per-bin displacement analyses, revealing an
835 exacerbated constrained chromosome rearrangements upon NapM loss.



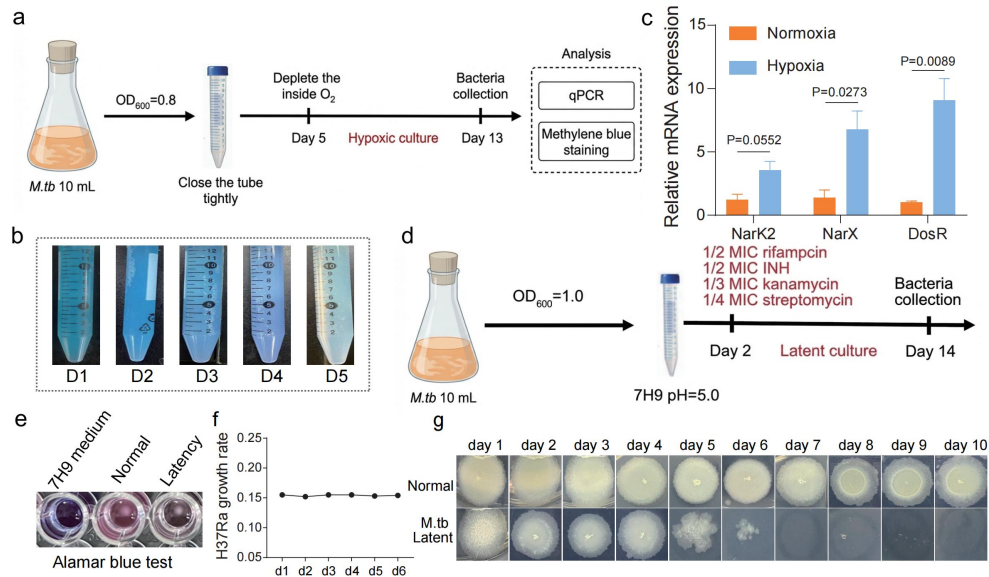
Extended Data Fig. 1 Experimental reproducibility and genomic features of the *M. tuberculosis* Hi-C dataset. **a**, Schematic overview of the experimental workflow for constructing chromosome Hi-C libraries in *M. tuberculosis*. **b**, Hi-C contact maps from four biological replicates with zoom-in views of representative regions, demonstrating high reproducibility. **c**, Scatter plots comparing pairwise correlations between replicates, confirming the consistency of chromosomal contact patterns. **d**, Genomic distribution of GC skew and GC content along the *M. tuberculosis* chromosome.



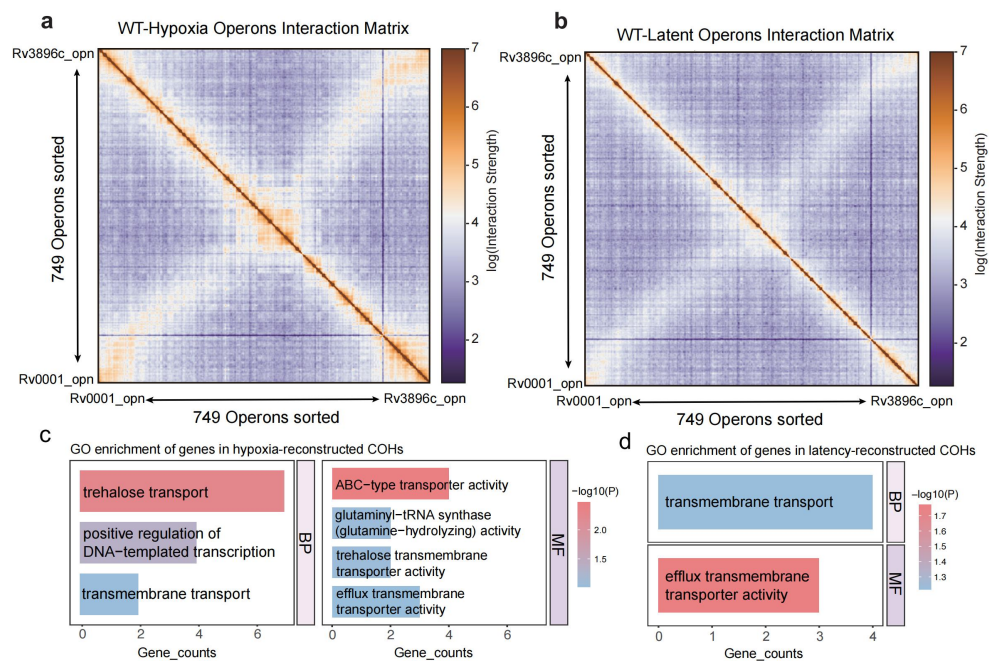
Extended Data Fig. 2 A central role of NapM(Rv0047c) in the multi-layer Protein DNA interaction network. **a**, Summary statistics of the integrated protein and DNA multi-layer interaction network, including the numbers of nodes and edges. **b**, Interaction network integrating COHs with binding proteins; **Rv0047c** (NapM) occupies a central position (red circle), linking multiple COHs to protein complexes. **c**, Differential Hi-C contact heatmap (*napM*-KO minus WT), highlighting widespread architectural alterations upon NapM depletion. **d**, Multiple sequence alignment of NapM homologs from *M. tuberculosis* and representative bacterial species (including *S. terrae*, *G. paraffinivorans*, *M. sp. KC723*, *B. subtilis*, and *S. criceti*), highlighting conserved residues and motifs across phylogenetically distant taxa. **e**, Broad conservation analysis of Rv0047c across bacteria, showing the taxonomic distribution of NapM homologs at the genus level, indicating widespread evolutionary conservation.



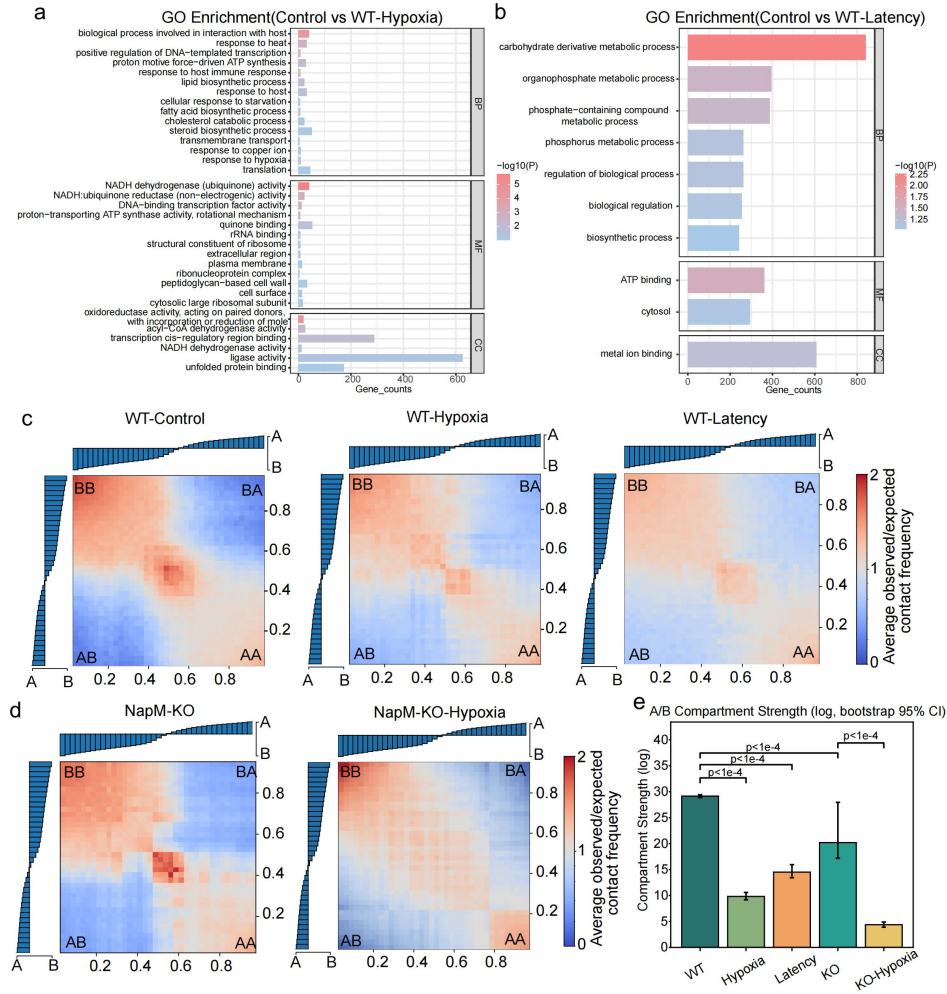
Extended Data Fig. 3 Integrated RNA-seq and ChIP-seq reveal NapM regulatory functions. **a**, RNA-seq volcano plot showing DEGs between *napM*-KO and WT strains. **b**, Comparison of chromosome interaction change amplitudes between DEG-DEG and Stable-Stable gene pairs. The y-axis represents the absolute $\log_2$ fold change of contact intensity between the KO and WT, and significance was assessed using a one-sided Mann-Whitney U test. **c**, Circular genome track visualization showing RNA-seq profiles for WT and *napM*-KO together with NapM ChIP-seq signal; COHs are annotated. **d**, Quadrant-based volcano plot integrating ChIP-seq and RNA-seq data. DEGs are classified according to the presence or absence of ChIP-seq peaks and their direction of regulation (up- or downregulated). Each quadrant indicates genes with ChIP-seq peaks or without peaks, with corresponding gene counts annotated. **e,f**, Representative loci illustrating the relationship between NapM binding and gene expression changes. **e**, *Rv2629*, which is bound by NapM, shows marked upregulation in the *napM*-KO strain. **f**, *Rv1195*, also associated with NapM binding, exhibits significant downregulation upon *napM* deletion.



Extended Data Fig. 4 Establishment and validation of hypoxic and latent models of *Mycobacterium tuberculosis*. **a**, Schematic illustration of the hypoxic culture model. Exponentially growing *M. tuberculosis* cultures ($OD_{600} = 0.8$) were transferred into sealed tubes to deplete intracellular oxygen and incubated under static conditions. Bacteria were collected at day 13 for downstream analyses, including qPCR and methylene blue staining. **b**, Methylene blue staining of cultures from day 1 to day 5 showing progressive decolorization, indicating oxygen depletion and establishment of hypoxia. **c**, Quantitative PCR analysis of hypoxia-responsive genes (*narK2*, *narX*, and *dosR*) under normoxic and hypoxic conditions. Data are shown as mean $\pm$ s.d. Statistical significance was determined by two-tailed Student's *t*-test; exact *P* values are indicated. **d**, Schematic illustration of the latent culture model. Cultures at $OD_{600} = 1.0$ were transferred into acidic 7H9 medium (pH 5.0) supplemented with sub-inhibitory concentrations of antibiotics (1/2 MIC rifampicin, 1/2 MIC isoniazid, 1/3 MIC kanamycin, and 1/4 MIC streptomycin) and incubated for 14 days before bacterial collection. **e**, Alamar Blue assay showing reduced metabolic activity in latent cultures compared with normal growth conditions and medium control. **f**, Growth rate analysis of *M. tuberculosis* H37Ra under normal and latent conditions, confirming growth arrest during latency induction. **g**, Spot assay performed over a 10-day period to assess bacterial culturability. Normal cultures continue to grow, whereas latent cultures progressively lose colony-forming ability, indicating entry into a dormant state.



Extended Data Fig. 5 Operon interactions are reorganized between WT hypoxia and latent conditions. **a**, Operon–operon interaction heatmap derived from Hi-C data under WT-hypoxia condition, showing pairwise contact frequencies among 749 annotated operons. **b**, Operon–operon interaction heatmap under WT-latent condition, illustrating the reorganization of interaction patterns among the same 749 operons during dormancy. **c,d**, GO enrichment analysis of genes within COHs reconstructed under hypoxic (**c**) and latent (**d**) conditions.



Extended Data Fig. 6 Functional enrichment and A/B compartment remodeling across stress conditions and NapM deficiency. **a**, Gene Ontology (GO) enrichment analysis of DEGs between wild-type control and WT-hypoxia conditions. Enriched terms are shown for the biological process (BP), molecular function (MF), and cellular component (CC) categories, highlighting suppression of translational machinery and respiratory-related processes during hypoxic adaptation. **b**, GO enrichment analysis of DEGs between wild-type control and WT-latent conditions. Enriched biological processes are predominantly associated with carbohydrate derivative metabolism, organophosphate metabolism, and biosynthetic processes, reflecting large-scale metabolic reprogramming during latency. **c**, Saddle plots of A/B compartment interactions derived from Hi-C data for WT-control, WT-hypoxia, and WT-latent conditions. Heatmaps represent observed/expected contact frequencies between genomic bins ranked by the first eigenvector (E1), illustrating progressive weakening of compartmental segregation under stress conditions. **d**, Saddle plots of A/B compartment interactions for *napM*-KO and *napM*-KO under hypoxic conditions. Loss of NapM results in altered compartmentalization patterns and reduced separation between A and B compartments, particularly under stress conditions. **e**, Quantification of A/B compartment strength across conditions, calculated from normalized saddle matrices and shown on a logarithmic scale. Values represent the mean compartment strength with 95% confidence intervals by bootstrap resampling. Statistical significance between groups was assessed using two-sided bootstrap tests; exact *p*-values are indicated, with $p < 1 \times 10^{-4}$ denoted where applicable.

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